



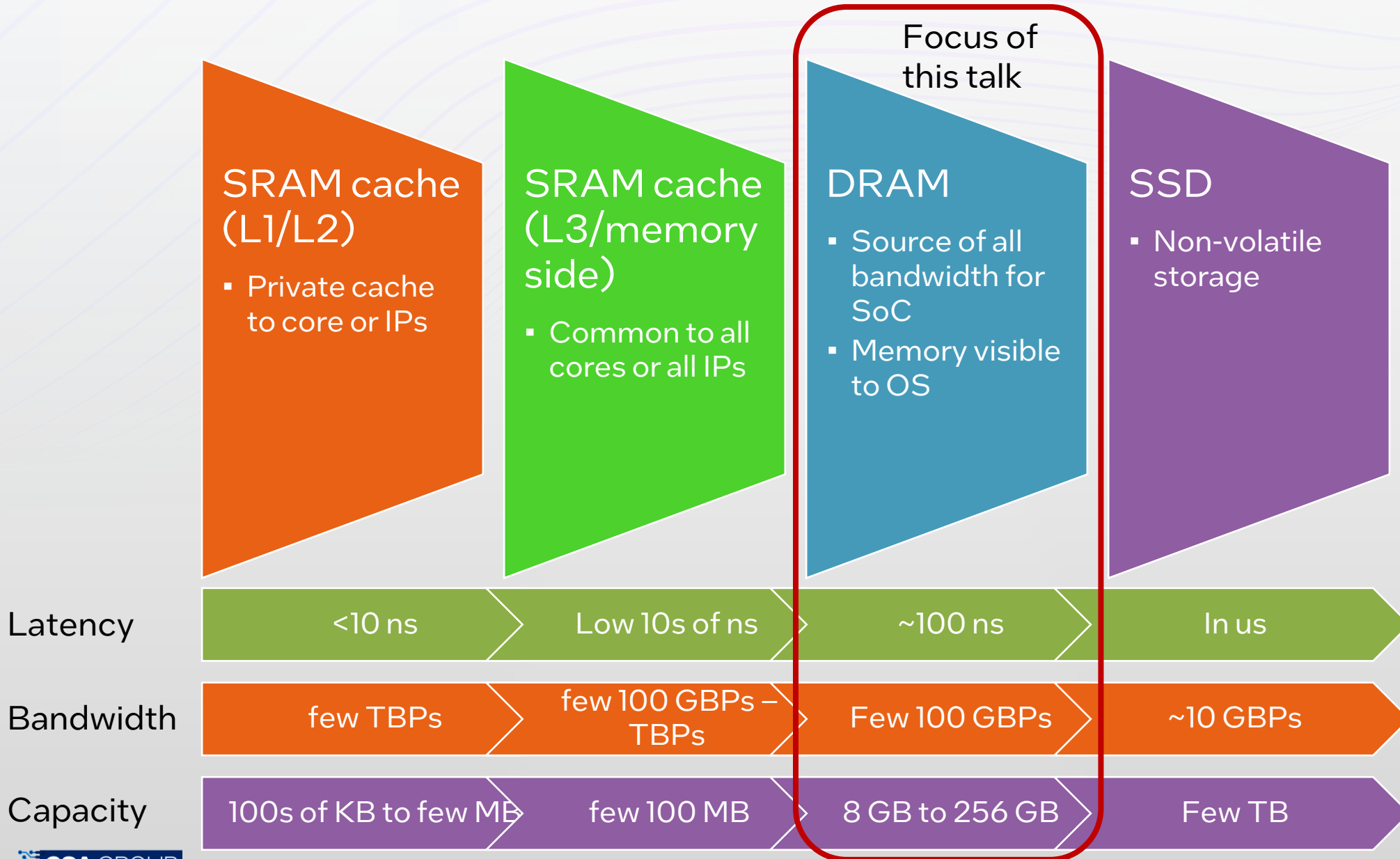
Challenges in scaling memory bandwidth

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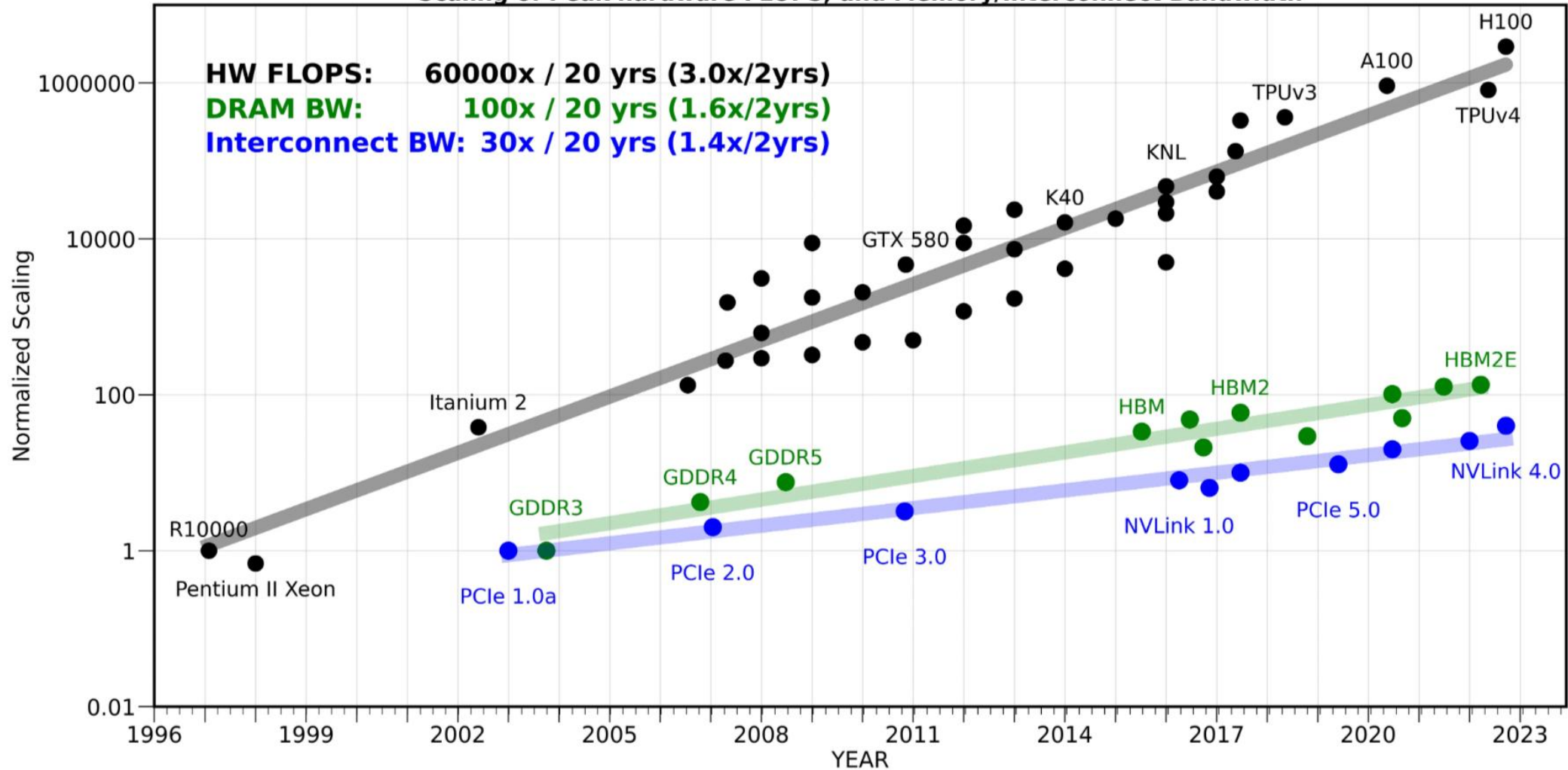


DRAM Memory in memory hierarchy



Memory wall

Scaling of Peak hardware FLOPS, and Memory/Interconnect Bandwidth



Memory bandwidth scaling is half of compute scaling even with high bandwidth memory options (GDDR/HBM)

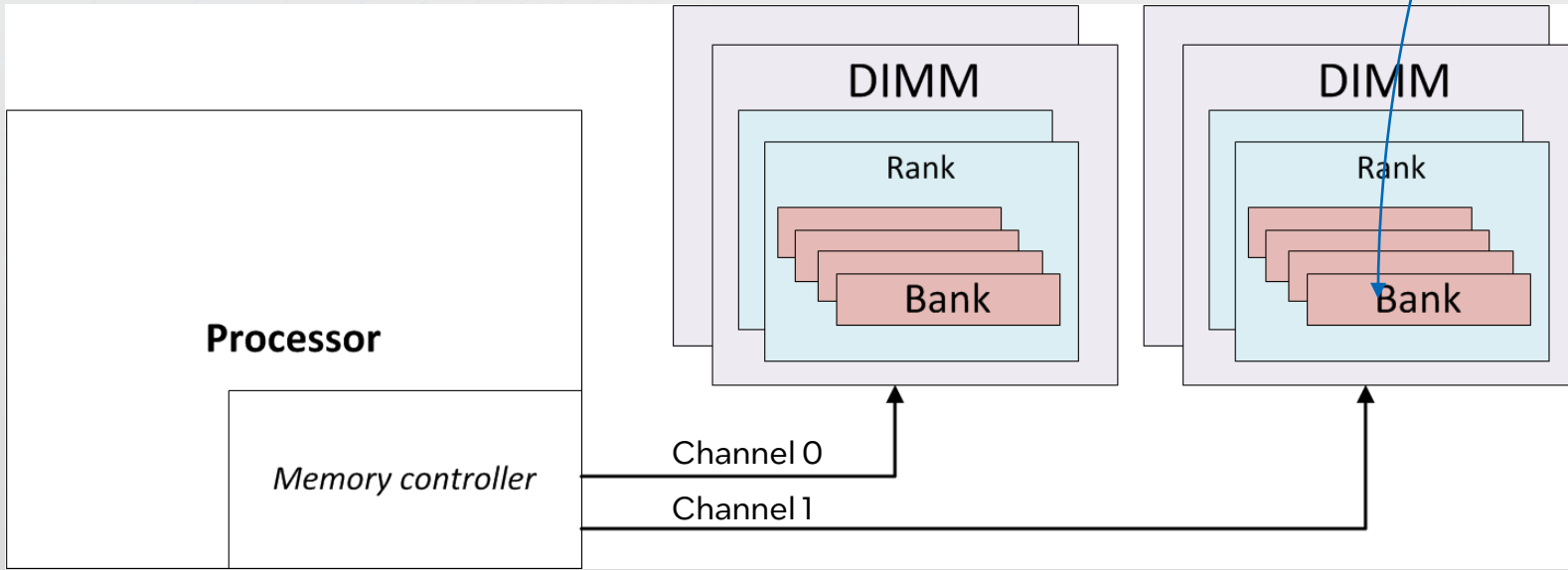
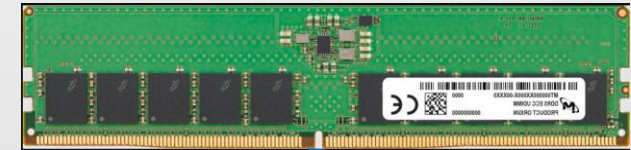
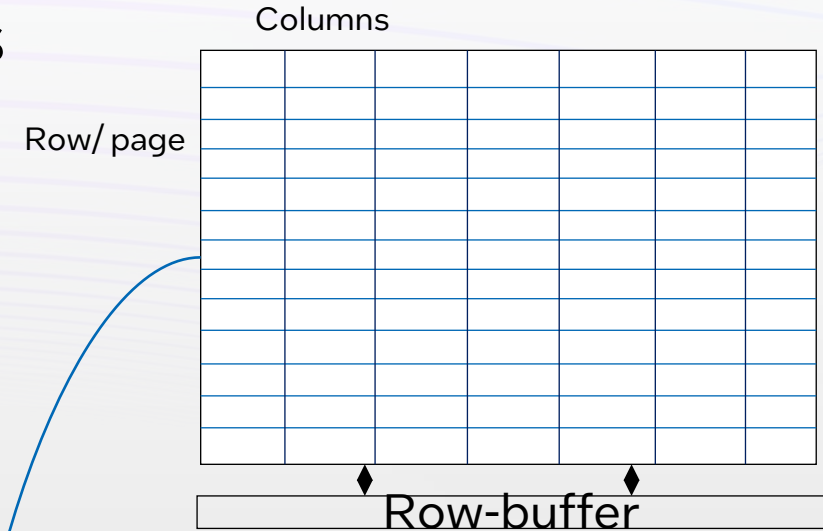
Different DRAM memory technologies

	DDR	LPDDR	GDDR	HBM
Used in	Desktops, laptops, servers	Smart phones, laptops	Discrete graphics cards	Discrete graphics cards, AI training SoCs
Optimized for	Performance, then power	Power, then performance	Performance	Performance and package area
Form factor	Removable/ soldered DIMMs	Soldered or Memory on Package	Soldered DIMMs	Memory on Package (2.5D)
Current generation	DDR5	LPDDR5x	GDDR7	HBM4
Bandwidth	90 GBPs for x128	120 GBPs for x128	448 GBPs for x128	1024 GBPs for x1024
Allow vertical stacking	Yes, only for capacity	No	No	Yes, for both capacity and bandwidth

All DRAM technologies use similar DRAM cell & organization and are standardized by JEDEC

DRAM Organization and operation basics

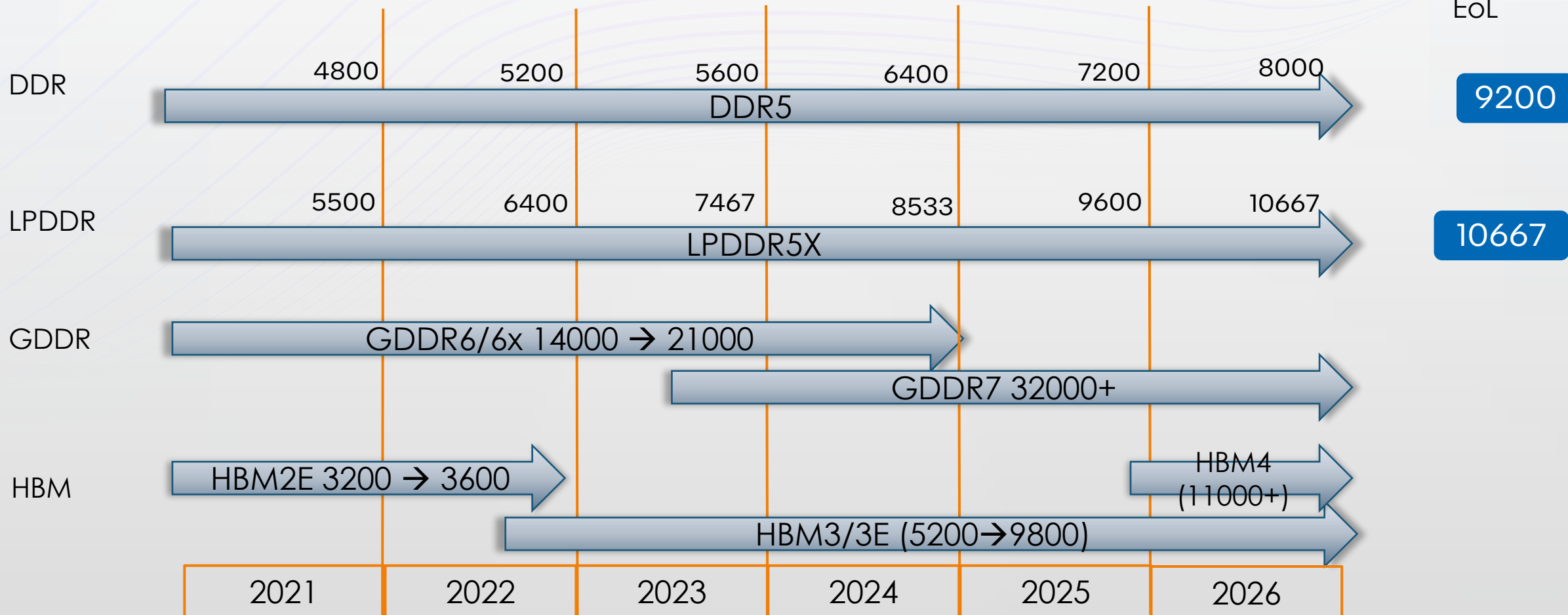
- DRAM organization is hierarchical to add capacity and bandwidth
- Smallest unit of parallel resource is a bank which is a matrix of rows and columns. Only one row can be open at a time in a bank
- Channels add capacity and bandwidth. But need more I/O
- Ranks and DIMMs add only capacity



DRAM command	Operation
ACT	Open row
CAS (Read/Write)	Read or Write to the open row
PRE	Close row

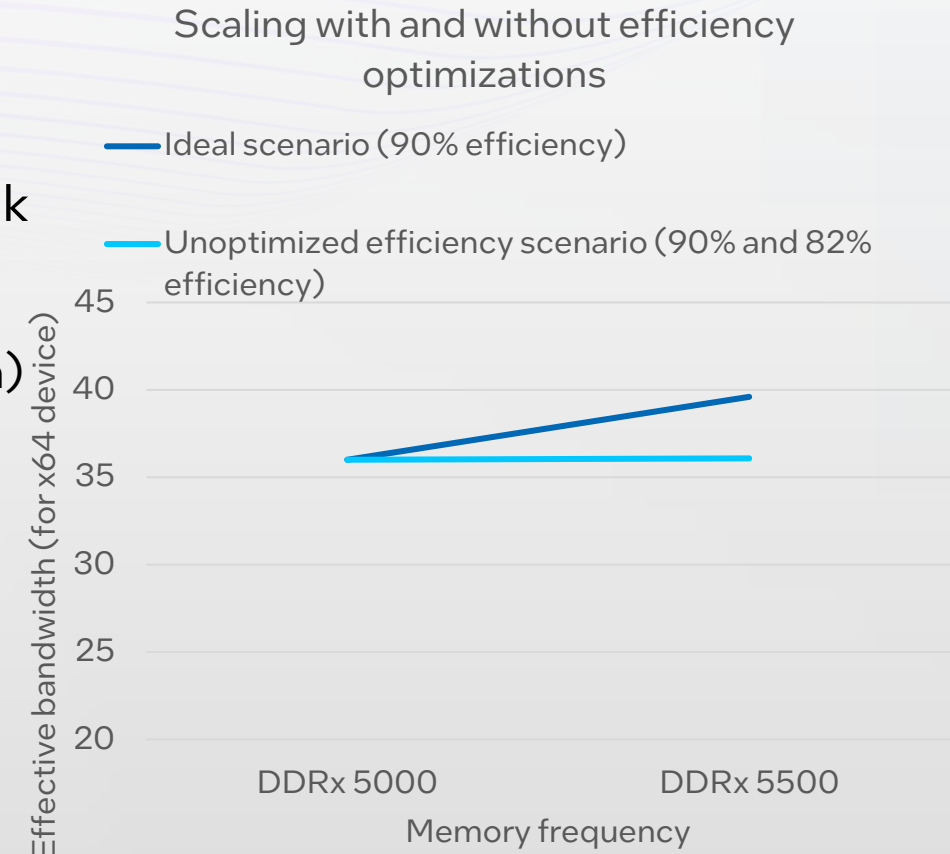
Each pair of commands to a given hierarchy has a timing constraint

Memory technology evolution



Memory bandwidth efficiency vs peak bandwidth

- Peak bandwidth is data rate * memory width
 - For instance, for DDR5 5600 2x64, bandwidth is 5.6 G * (2*64/8) = 89.6 GBPs
- Memory bandwidth efficiency = delivered bandwidth/ peak bandwidth
 - Applies to only cases where memory bandwidth is fully saturated (otherwise this term is called bandwidth utilization)
- Delivering high efficiency is not straightforward in DRAM
 - Need to obey DRAM timings between each operation/ hierarchy
 - Can cause very low efficiencies when done in unoptimized manner
 - Requests are scheduled in an optimized manner by Memory controller (MC) in the SoC
 - Need certain parallelism and locality from workloads and some resources from DRAM device for delivering high bandwidth



Delivering good bandwidth efficiency is a combined responsibility of SoC, DRAM device architecture and workloads

This talk will untangle and spell out the components of DRAM bandwidth efficiency

Challenges in scaling memory bandwidth and density

- Power/ thermals
 - Typically, each DRAM generation lowers operating voltage to reduce power consumption
 - This lowers mW per GBps but overall power does increase due to GBps increase
 - This can cause Bandwidth throttling in some memory types (like DDR5)
 - Especially true for consumer memory with no active cooling
- Reliability
 - Probability of Soft errors increase with higher memory densities
 - Current DRAM includes provisions for Error detection/ correction at 3 different levels
 - Soft error in DRAM
 - Link error (during data transfer on DDR bus)
 - End2end system level error handling
- Security
 - Security vulnerabilities increase with higher memory densities
 - Row hammer [[Kim et al ISCA 2014](#)]- data can be flipped in adjacent rows due to repeated activates in a target row
 - Row press [[Luo et al ISCA 2023](#)]- similar effect for keeping a row open for longer duration
- Memory bandwidth efficiency
 - Due to above reasons and other structural reasons, it is challenging to maintain high bandwidth efficiency when scaling memory frequency

Learn why/ how for following

- What happens to memory bandwidth efficiency when dram frequency increases in the same generation?
 - Increases/ **Decreases**/ Stays flat
- What happens to memory bandwidth efficiency with process node/ density advancement?
 - Increases/ **Decreases**/ Stays flat
- Access locality and parallelism are both important in a workload. Given a choice which do we prefer?
 - **Access locality**/ parallelism
- Memory bandwidth efficiency improvements are best done
 - With in Memory Controller/ **end to end SoC & software**
- Finding most bandwidth optimized schedule for a given set of n requests in Memory controller is a
 - Known algorithm/ **NP-hard problem**

Analytical model for DRAM efficiency

- $$\text{DRAM efficiency} = \min \left(\frac{RLP * CAS \text{ per ACT} * tBL}{tRRD}, \frac{BLP * CAS \text{ per ACT} * tBL}{tRC}, \frac{CAS \text{ per ACT} * tBL}{(CAS \text{ per ACT} * cmdlength_{CAS}) + cmdlength_{ACT} + cmdlength_{PRE}} \right)$$
- $$= \min \left(\frac{RLP * CAS \text{ per ACT} * tBL}{tRRD}, \frac{BLP * CAS \text{ per ACT} * tBL}{tRC}, \text{command bandwidth constraint} \right)$$

These constraints are captured in an analytical model for memory efficiency used internally at Intel

Timing parameters

$$= \min \left(\frac{RLP * CAS \text{ per ACT} * tBL}{tRRD}, \frac{BLP * CAS \text{ per ACT} * tBL}{tRC}, \text{command bandwidth constraint} \right)$$

- tRC → Row cycle time - Minimum time between two row opens (ACTs) in same bank
- tRRD → minimum time between two row opens (ACTs) in same rank
- Read tRC has been fixed at 60 ns for LP technologies and 45 ns for DDR technologies
 - Fixed in ns → increases linearly in cycles with frequency → **hurts bandwidth efficiency**
- Write tRC in ns saw a **step increase** in DDR5/LP5 because of DRAM ECC
- Read and Write tRC in ns will see **effective increases** in LPDDR6/DDR6 because of Row hammer mitigation

Timing parameters increase linearly or more than linearly with increasing frequency
→ Need to increase factors in the numerator to keep bandwidth efficiency flat

Parallelism

$$= \min \left(\frac{RLP * CAS \text{ per ACT} * tBL}{tRRD}, \frac{BLP * CAS \text{ per ACT} * tBL}{tRC}, \text{command bandwidth constraint} \right)$$

How to increase parallelism (both Bank level and Rank level)?

- Higher number of banks and ranks
 - Banks → has been saturating at 16 for LPDDR and 32 for DDR due to cost reasons
 - Ranks → Ranks automatically add capacity; hence memory cost also increases linearly
- Memory controller queue depth
 - Higher queue depth probabilistically allows more BLP for the same number of banks
 - Queue depths may need to be increased to maintain bandwidth scaling
 - Innovative micro-arch schemes can also increase effective queue depth while keeping same physical queues
- Workload behavior and DRAM address map
 - Workload behavior and DRAM address map needs to be tuned together to enable high parallelism

	banks
LPDDR3	8
LPDDR4	16
LPDDR5	16
LPDDR6	16

MC can increase parallelism significantly but there are hard limits to this

Locality

$$= \min \left(\frac{RLP * \text{CAS per ACT} * tBL}{tRRD}, \frac{BLP * \text{CAS per ACT} * tBL}{tRC}, \text{command bandwidth constraint} \right)$$

- CAS per ACT → Number of cache line accesses for every row open
- Streaming workloads have more locality and hence mostly have better bandwidth efficiency
- General purpose, multi core x86 workloads tend to have less locality
- How to improve CAS per ACT?
 - MC micro-arch plays a minor role – deeper queues help in higher CAS per ACT
 - SoC micro-arch also impacts CAS per ACT – need to minimize re-ordering of requests coming from compute IPs to keep the workload CAS per ACT intact till it reaches MC
 - Workload behavior dominates CAS per ACT (along with address map tuning)

Workload and SoC Arch play key role in improving CAS per ACT

Burst length/ Channel width

$$= \min \left(\frac{RLP * CAS \text{ per ACT} * tBL}{tRRD}, \frac{BLP * CAS \text{ per ACT} * tBL}{tRC}, \text{command bandwidth constraint} \right)$$

- Burst length → number of cycles to transfer a cache line → inversely proportional to channel width
- Example – DDR5 is x32. Burst length = 512b/ 32 = 16
- Narrow channels increase bandwidth efficiency
 - Primary knob for DRAM technology evolution
- What is the trade off with narrow channels?
 - Need more controllers – more design resources
 - Increases cache line latency (though higher frequency balances this)
 - Indirectly hurts parallelism and locality – workload has limited parallelism and locality and dividing it among more controllers reduces it

	Burst length
LPDDR3	8
LPDDR4	32
LPDDR5	32
LPDDR6	48

Increasing burst length has been main way to keep flat efficiency with technology transition
But this will also hit hard limits soon - LP6 channels are only 12 bits wide

$$\frac{CAS\ per\ ACT * tBL}{(CAS\ per\ ACT * cmdlength_{CAS}) + cmdlength_{ACT} + cmdlength_{PRE}}$$

Command bandwidth constraint

- At extreme point of 1 CAS per ACT (random access) command bandwidth constraint is possible when
 - Sum of effective command length of ACT, CAS, PRE commands > burst length
 - Effective command length is a function of native command length, MC uarch and electrical constraints (aka n mode)
- Command bus needs to be narrow especially for narrow channels to keep down SoC IO pin count → increase in native command length
- MC uarch: Gearing (frequency ratio) is a common SoC method to keep MC operating at lower frequencies compared to DDR
 - For instance, Intel ARL refresh supports up to DDR5 7200 in gear 2 (2:1) and beyond that in gear 4 (4:1)
 - When operating at higher gears, MC issuing only 1 command per cycle may cause command bandwidth limitation
- SoC improvements to avoid command bandwidth constraint
 - Keep lower gears or issue multiple commands per cycle from MC
 - Use combo commands like Auto precharge with good heuristics

SoC has limited influence over command bandwidth constraint –
best way is to improve workload locality

Summary

- It is challenging to keep bandwidth efficiency flat with increasing DRAM frequency
 - Reliability, Security and above all device physics works against flat efficiency
 - DRAM technology generational improvements help in this but often not enough
 - MC uarch optimizations play key role in improving bandwidth efficiency
 - End to end SoC and workload optimizations are also required to reach the objective – optimizing only MC results in more brute force solutions which affect PPA (Power-Perf-Area) metric
- This talk covered only basics of bandwidth efficiency and focused on read only/ write only bandwidth efficiency. Read-write bandwidth efficiency includes more constraints
- Watch out for an expanded set of talks which will cover this topic from fundamental DRAM cell organization in a follow up 1 credit course